National Tsing Hua University
1130IEEM 513600
Deep Learning and Industrial Applications **Homework 2**

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1. (20 pts) Select 2 hyper-parameters of the artificial neural network used in Lab 2 and set 3 different values for each. Perform experiments to compare the effects of varying these hyper-parameters on the loss and accuracy metrics across the training, validation, and test datasets. Present your findings with appropriate tables.

2. (20 pts) Based on your experiments in Question 1, analyze the outcomes. What differences do you observe with the changes in hyper-parameters? Discuss whether these adjustments contributed to improvements in model performance, you can use plots to support your points. (Approximately 100 words.)

3. (20 pts) In Lab 2, you may have noticed a discrepancy in accuracy between the training and test datasets. What do you think causes this occurrence? Discuss potential reasons for the gap in accuracy. (Approximately 100 words.)

4. (20 pts) Discuss methodologies for selecting relevant features in a tabular dataset for machine learning models. Highlight the importance of feature selection and how it can impact model performance. You are encouraged to consult external resources to support your arguments. Please cite any sources you refer to. (Approximately 100 words, , excluding reference.)

5. (20 pts) While artificial neural networks (ANNs) are versatile, they may not always be the most efficient choice for handling tabular data. Identify and describe an alternative deep learning model that is better suited for tabular datasets. Explain the rationale behind its design specifically for tabular data, including its key features and advantages. Ensure you to reference any external sources you consult. (Approximately 150 words, excluding reference.)

Answer:

1. **Epoch Table**:

	Epoch: 50	**Epoch: 200**	**Epoch: 500**
Train loss	0.4576	0.2370	0.1895
Train acc	78.8360%	90.4762%	94.7090%
Val loss	0.5193	0.6311	0.5828
Val acc	75.3086%	71.6049%	81.4815%
Best Val loss	0.5125	0.5632	0.3982
Best Val acc	75.31%	76.54%	83.95%
Test accuracy	70.96774193548387%	67.74193548387096%	77.41935483870968%

As epochs increase (50 to 500), train loss decreases (0.4576 to 0.1895) and train accuracy rises (78.84% to 94.71%), showing better training fit. Validation loss fluctuates (0.5193 to 0.5828), but best validation accuracy improves (75.31% to 83.95%). Test accuracy also rises (70.97% to 77.42%), suggesting some generalization improvement.

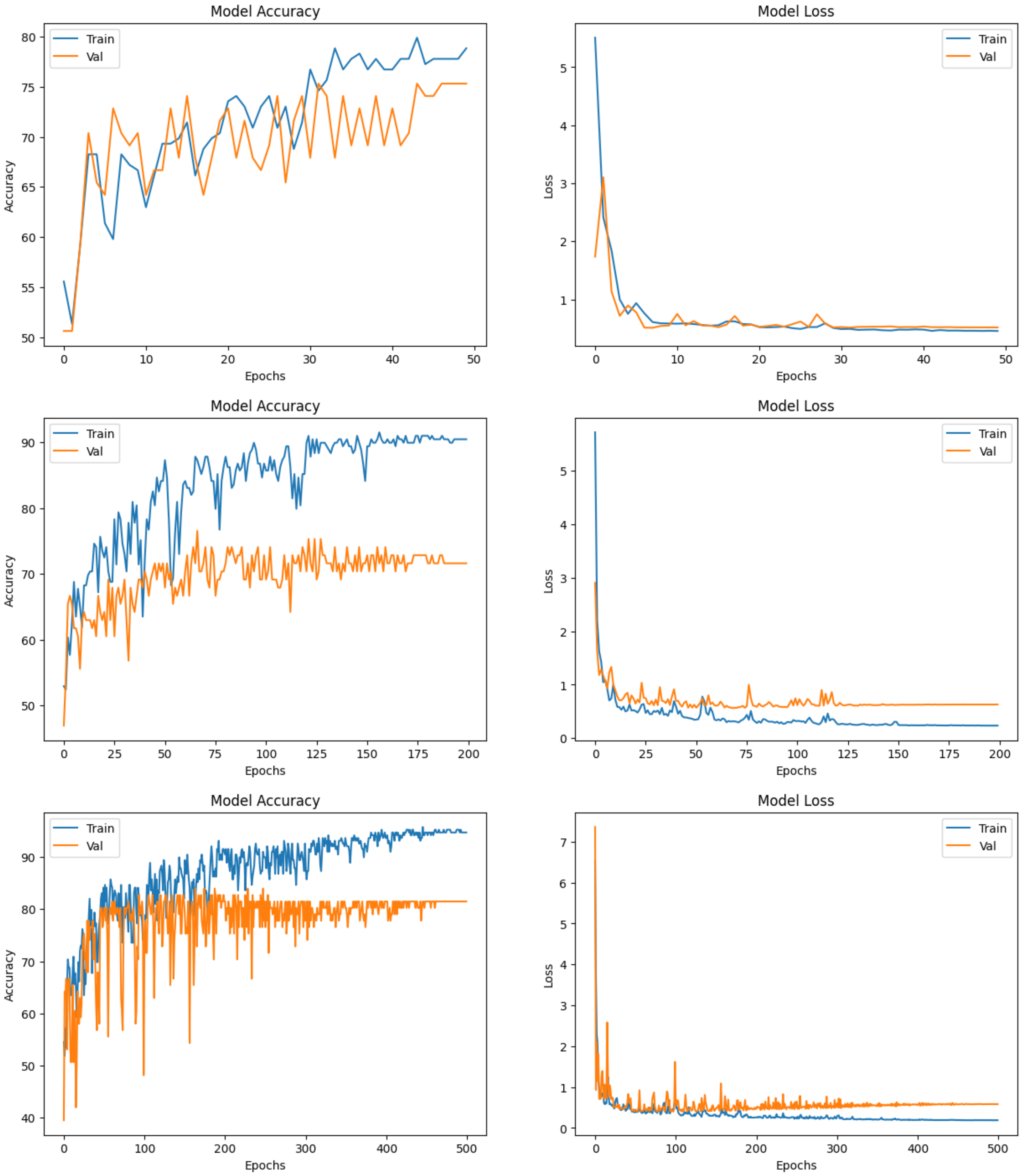
LR Table:

	LR: 1e-2	**LR: 1e-4**	**LR: 1e-6**
Train loss	0.2662	0.4944	0.9349
Train acc	88.3598%	76.7196%	55.5556%
Val loss	0.5402	0.5550	1.0595
Val acc	79.0123%	71.6049%	51.8519%
Best Val loss	0.5062	0.5518	51.8519%
Best Val acc	80.25%	74.07%	51.85%
Test accuracy	80.64516129032258%	64.51612903225806%	48.38709677419355%

At LR 1e-2, train loss is low (0.2662) and accuracy high (88.36%), with test accuracy peaking at 80.65%. At LR 1e-4, performance drops (train acc 76.72%, test acc 64.52%). At LR 1e-6, results worsen significantly (train acc 55.56%, test acc 48.39%), indicating underfitting due to a tiny learning rate.

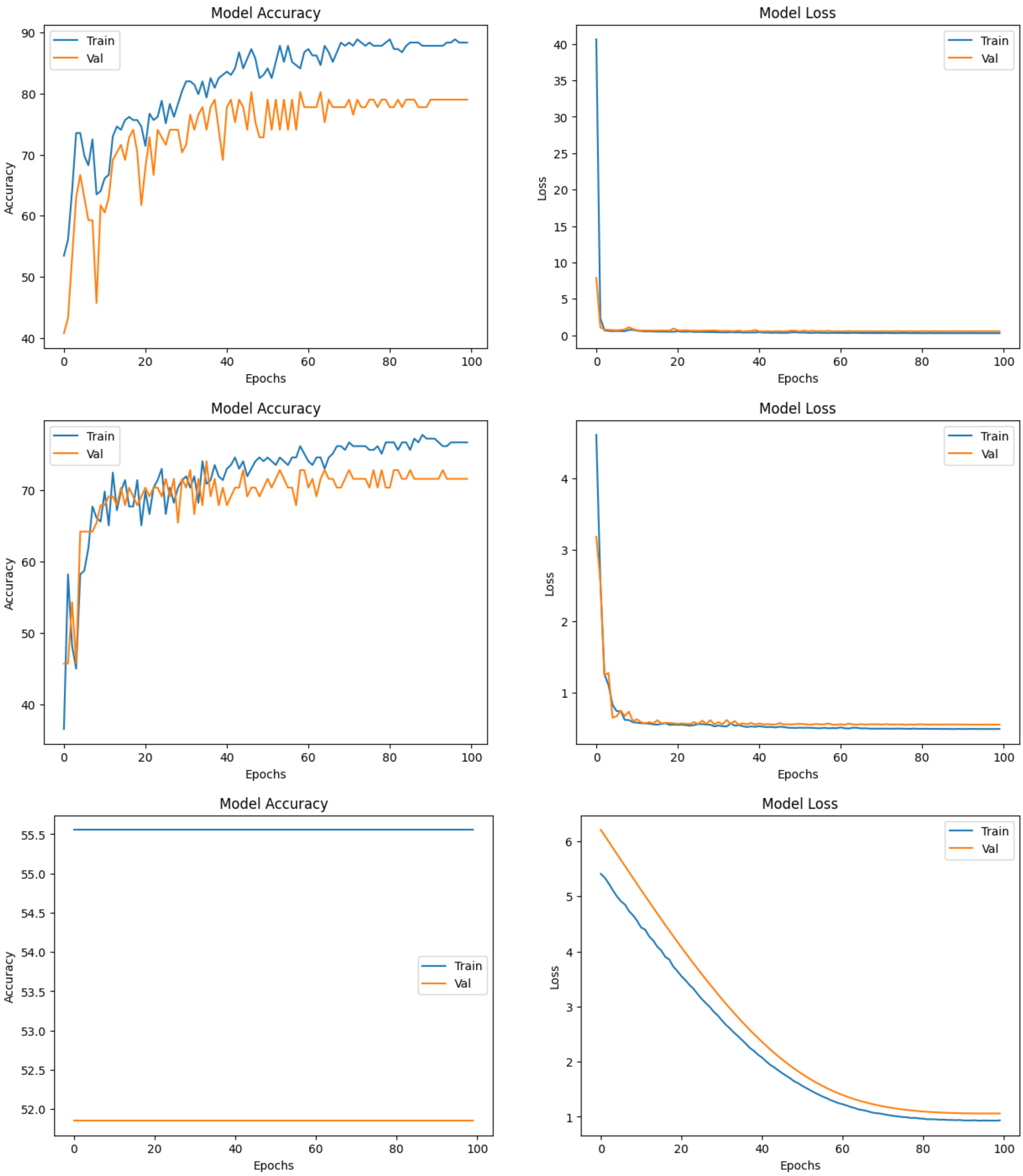
2. Higher epochs improve training fit and slightly boost test accuracy (77.42% at 500 vs. 70.97% at 50), but validation loss suggests potential overfitting.

Graph of the different epochs(50, 200, 500):



Learning rate has a stronger effect: 1e-2 yields the best balance (test acc 80.65%), while 1e-6 cripples learning (test acc 48.39%).

Graph of the different learning rate(1e-2, 1e-4, 1e-6):



Adjustments like increasing epochs or optimizing LR improve performance, but extreme LR values degrade it.

3. The significant accuracy gap between training and test sets indicates overfitting. The model excels on training data (94.71% vs. 77.42%) but performs poorly on unseen data. This likely stems from the model memorizing noise, not genuine patterns. Insufficient regularization, high complexity, or excessive training epochs contribute. Data distribution differences between sets could also be a factor. To address this, consider adding dropout or weight decay, simplifying the model, reducing epochs, or adjusting the learning rate (1e−2) to improve generalization.

4. Feature selection is vital for tabular data in machine learning. Techniques like filter, wrapper, and embedded methods identify key features, improving model performance. By reducing dimensionality and noise, selection prevents overfitting and enhances interpretability. Removing irrelevant features, as shown in studies like Guyon & Elisseeff (2003), can increase accuracy and decrease computational costs. Aligning features with the target variable directly improves predictive power, making it a crucial step in model development.

Reference: Guyon, I., & Elisseeff, A. (2003). An introduction to variable and feature selection. *Journal of Machine Learning Research*, 3, 1157-1182.
https://www.jmlr.org/papers/volume3/guyon03a/guyon03a.pdf

5. Artificial Neural Networks (ANNs), despite their success with unstructured data, often face challenges with tabular datasets, including overfitting and inefficiency. TabNet, introduced by Arik & Pfister (2021), offers a tailored solution. It employs a sequential attention mechanism, selecting relevant features at each decision step, mirroring the interpretability of decision trees. TabNet's sparse feature selection minimizes noise influence, while its transformer-like architecture is optimized for structured data. This results in improved generalization compared to standard ANNs, particularly on smaller datasets, and inherent interpretability, crucial in domains like finance and healthcare. TabNet's performance on benchmarks, such as the UCI dataset, demonstrates its superiority over traditional ANNs in tabular data scenarios.

Reference: Arik, S. Ö., & Pfister, T. (2021). TabNet: Attentive interpretable tabular learning. *Proceedings of the AAAI Conference on Artificial Intelligence*, 35(8), 6679-6687.
https://arxiv.org/pdf/1908.07442